SCHOOL OF COMPUTING AND INFORMATICS | DEPARTMENT OF COMPUTER SCIENCE

Individual Assignment

CCS 3354: Data Mining — Weeks 1–6 | Total: 100 Marks

General Instructions

1. This assignment covers Weeks 1 to 6: Introduction to Data Mining, Data Warehousing, Data Preprocessing (Parts 1 & 2), Data Mining Primitives & Query Languages, and Association Rule Mining.
2. Answer ALL questions in all sections. Read each question carefully before responding.
3. Show all working where calculations are involved. Unsupported answers will not attract full marks.
4. Write clearly and concisely. Where definitions are required, use precise technical language.
5. Plagiarism will result in zero marks. This is an individual assignment — collaboration is not permitted.
6. Submit a printed hard copy AND an electronic copy via the LMS by the stipulated deadline.
7. Late submissions will be penalised at 10 marks per day unless prior written approval is obtained.

Student Declaration & Details

Student Name:	
Registration No.:	
Programme:	Bachelor of Computer Science
Course Code:	CCS 3354
Course Title:	Data Mining
Lecture/Group:	
Submission Date:	

Declaration: I confirm that this assignment is entirely my own work. I have not copied from any other student or source without proper citation, and I understand the consequences of academic dishonesty.

Signature _____ Date _____

Marks Allocation Summary

Section	Topic Covered	Max Marks	Marks Obtained	Examiner
Section A	Introduction to Data Mining & Data Warehousing (Weeks 1–2)	**20**		
Section B	Data Preprocessing — Parts 1 & 2 (Weeks 3–4)	**25**		
Section C	Data Mining Primitives & Query Languages (Week 5)	**20**		
Section D	Association Rule Mining — Part 1 (Week 6)	**25**		
Section E	Integrative / Applied Case Study (Weeks 1–6)	**10**		
TOTAL	All Sections	**100**		

SECTION A — Introduction to Data Mining & Data Warehousing [20 Marks]

This section tests your understanding of core data mining concepts and data warehouse fundamentals from Weeks 1 and 2.

Q1 — Knowledge Discovery in Databases (KDD) and Data Mining Fundamentals

Answer all four parts below.

10 marks

(a)	Define data mining and distinguish it from traditional database querying. In your answer, explain why data mining is necessary in modern data-driven organisations.	**(3 marks)**
(b)	Describe FOUR major functionalities of a data mining system, providing one real-world example for each functionality.	**(4 marks)**
(c)	Outline the six key steps of the KDD (Knowledge Discovery in Databases) process and briefly describe what happens at each step.	**(3 marks)**

Q2	**Data Warehousing Concepts and OLAP Operations** Answer all four parts below.	10 marks

(a)	Explain the fundamental architectural difference between an OLTP (Online Transaction Processing) system and an OLAP (Online Analytical Processing) system. Use a concrete example to illustrate each.	**(3 marks)**
(b)	A retail company has a data warehouse structured around sales transactions. Draw and label a Star Schema for this scenario, identifying the fact table and at least four dimension tables.	**(4 marks)**
(c)	Briefly explain THREE of the following OLAP cube operations, providing an example of each: (i) Roll-up, (ii) Drill-down, (iii) Slice, (iv) Dice.	**(3 marks)**

SECTION B — Data Preprocessing: Parts 1 & 2 [25 Marks]

This section covers data quality, cleaning, transformation, discretisation, and dimensionality reduction from Weeks 3 and 4.

Q3	**Data Quality and Cleaning** Answer all three parts below.	10 marks

(a)	Identify and explain FOUR dimensions of data quality. For each dimension, provide an example of a problem that could arise and a corresponding remedy.	**(4 marks)**
(b)	A student dataset contains the following age values: {22, 24, 21, 23, 150, 25, 22}. Identify the anomaly, name the technique you would use to detect it, and describe TWO methods for treating it.	**(3 marks)**
(c)	Distinguish between data integration and data transformation. Describe the entity identification problem that arises during data integration and explain how it is resolved.	**(3 marks)**

Q4	**Data Transformation, Normalisation & Dimensionality Reduction** Answer all four parts below.	15 marks

(a)	The following dataset records students' exam scores: {45, 72, 88, 56, 91, 63, 77}. Apply Min-Max normalisation to map all values to the range [0, 1]. Show all working.	**(4 marks)**
(b)	Using the same dataset as in (a), apply Z-Score (standardisation) normalisation. Compute the mean and standard deviation, then normalise each value. Show all working.	**(4 marks)**
(c)	Explain the concept of the 'Curse of Dimensionality' and describe why it is problematic for data mining algorithms. List FOUR techniques used to address this problem.	**(4 marks)**
(d)	Describe the five steps of Principal Component Analysis (PCA) and explain what is meant by 'variance explained' by a principal component. Under what circumstances might PCA be inappropriate?	**(3 marks)**

SECTION C — Data Mining Primitives & Query Languages [20 Marks]

This section tests your knowledge of the building blocks of data mining tasks, DMQL, system architectures, and concept description from Week 5.

Q5	**Data Mining Primitives and DMQL** Answer all three parts below.	10 marks

(a)	List and explain ALL FIVE data mining primitives, describing how each one contributes to specifying a complete data mining task. Use a cohesive example throughout (e.g., a banking scenario).	**(5 marks)**
(b)	Write a valid DMQL query that mines association rules from a supermarket database called 'SuperMartDB'. Your query should reference a location hierarchy for customers, target the items and sales tables, and apply thresholds of min_support = 3% and min_confidence = 60%. Annotate each clause.	**(3 marks)**
(c)	Explain the role of concept hierarchies in data mining. Provide two examples of concept hierarchies from different domains (e.g., time, geography, product category).	**(2 marks)**

Q6	**Data Mining System Architectures & Concept Description** Answer both parts below.	10 marks

(a)	Compare and contrast the FOUR levels of coupling between a data mining system and a database/data warehouse: No Coupling, Loose Coupling, Semi-Tight Coupling, and Tight Coupling. For each level, describe one practical advantage and one limitation. Present your answer in a structured manner.	**(6 marks)**
(b)	Distinguish between Data Characterisation and Data Discrimination (Comparison) as forms of concept description. Provide a clearly worked example of each using a dataset of your choice — the example must include the type of output produced (e.g., chart type, rule form).	**(4 marks)**

SECTION D — Association Rule Mining: Part 1 [25 Marks]

This section tests support, confidence, lift, market basket analysis, and the Apriori algorithm from Week 6.

Q7 — **10 marks**

Support, Confidence, and Lift

Consider the following transaction database (N = 10 transactions):

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T1: {A, B, C}   T2: {A, C}   T3: {A, B, D}   T4: {B, C, E}   T5: {A, B, C, D}
T6: {B, C}   T7: {A, D}   T8: {A, B, C}   T9: {C, D, E}   T10: {A, B}
```

Use min_support = 50% and min_confidence = 70% for parts (a) to (c).

(a)	Calculate the support for each of the following itemsets: {A}, {B}, {C}, {A, B}, {A, C}, {B, C}, and {A, B, C}. Clearly show your working for each calculation.	**(4 marks)**
(b)	Determine which of the following rules are strong (satisfy both min_support and min_confidence): (i) {A} => {B}, (ii) {B} => {C}, (iii) {A, B} => {C}. Show all support and confidence calculations.	**(4 marks)**
(c)	Calculate the Lift for each rule identified as strong in part (b). Interpret each lift value — what does it reveal about the relationship between the items?	**(2 marks)**

Q8	**Apriori Algorithm & Market Basket Analysis** Answer all four parts below.	15 marks

(a)	State the Apriori property (anti-monotonicity principle) formally and informally. Explain why this property is critical to the efficiency of the Apriori algorithm and give a concrete example of how it enables candidate pruning.	**(4 marks)**
(b)	Using the transaction database from Question 7 with min_support = 50%, manually execute the Apriori algorithm. List: (i) all candidate 1-itemsets C1 and frequent 1-itemsets L1, (ii) all candidate 2-itemsets C2 (showing pruned candidates) and frequent 2-itemsets L2, (iii) all candidate 3-itemsets C3 and frequent 3-itemsets L3.	**(6 marks)**
(c)	A large supermarket chain wants to use market basket analysis to improve sales. Describe THREE specific, actionable business decisions the supermarket could make based on discovered association rules. For each decision, explain which stakeholder would benefit and how.	**(3 marks)**
(d)	The Apriori algorithm has known performance limitations. Describe TWO specific optimisation strategies (e.g., hash-based methods, partitioning) and explain how each reduces the computational cost of frequent itemset mining.	**(2 marks)**

SECTION E — Integrative Case Study [10 Marks]

This section requires you to apply and synthesise knowledge from across all six weeks. Read the scenario carefully before answering.

Q9 — **10 marks**

Case Study: National Health Monitoring Initiative

The Ministry of Health has commissioned a data mining project to analyse patient records from the past five years across 47 county hospitals. The dataset contains 2.3 million records with 120 attributes including: patient demographics, diagnoses (ICD-10 codes), lab results, medication prescriptions, length of stay, readmission rates, and geographic information. The data is sourced from disparate hospital information systems with varying formats, and is known to contain missing values, duplicate records, and inconsistent coding conventions.

Answer all three parts below.

(a)	Data Preparation Plan: As the lead data scientist, outline a comprehensive data preprocessing pipeline for this dataset. Your plan must address: (i) three specific data quality problems likely to exist in this dataset and how you would resolve each, (ii) which normalisation technique you would apply to numerical lab results and why, (iii) how you would reduce the 120-attribute dataset to the most relevant features before mining. Justify all choices.	**(4 marks)**
(b)	Mining Strategy: The Ministry wants to discover which combinations of conditions, demographics, and medications are associated with 30-day hospital readmission. Specify a complete data mining task using the five primitives (task-relevant data, knowledge type, background knowledge, interestingness measures, and presentation format). Then write a DMQL query implementing your specification.	**(4 marks)**
(c)	Critical Evaluation: Identify TWO ethical concerns and ONE technical limitation specific to mining patient health records for association patterns. For each concern, propose a concrete mitigation strategy.	**(2 marks)**

— END OF ASSIGNMENT —

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